Another way of posing the problem is to ask oneself: what is the ‘present’? We say that only the things of the present exist: the past no longer exists and the future doesn’t exist yet. But in physics there is nothing that corresponds to the notion of the ‘now’. Compare ‘now’ with ‘here’. ‘Here’ designates the place where a speaker is: for two different people ‘here’ points to two different places. Consequently ‘here’ is a word the meaning of which depends on where it is spoken. The technical term for this kind of utterance is ‘indexical’. ‘Now’ also points to the instant in which the word is uttered, and is also classed as ‘indexical’. But no one would dream of saying that things ‘here’ exist, whereas things which are not ‘here’ do not exist. So then why do we say that things that are ‘now’ exist and that everything else doesn’t? Is the present something which is objective in the world, that ‘flows’ and that makes things ‘exist’ one after the other, or is it only subjective, like ‘here’?

This may seem like an abstruse mental problem. But modern physics has made it into a burning issue, since special relativity has shown that the notion of the ‘present’ is also subjective. Physicists and philosophers have come to the conclusion that the idea of a present that is common to the whole universe is an illusion, and that the universal ‘flow’ of time is a generalization that doesn’t work. When his great Italian friend Michele Besso died, Einstein wrote a moving letter to Michele’s sister: ‘Michele has left this strange world a little before me. This means nothing. People like us, who believe in physics, know that the distinction made between past, present and future is nothing more than a persistent, stubborn illusion.’

Illusion or not, what explains the fact that for us time ‘runs’, ‘flows’, ‘passes’? The passage of time is obvious to us all: our thoughts and our speech exist in time; the very structure of our language requires time – a thing ‘is’ or ‘was’ or ‘will be’. It is possible to imagine a world without colours, without matter, even without space, but it’s difficult to imagine one without time. The German philosopher Martin Heidegger emphasized our ‘dwelling in time’. Is it possible that the flow of time which Heidegger treats as primal is absent from descriptions of the world?

Some philosophers, the most devoted followers of Heidegger among them, conclude that physics is incapable of describing the most fundamental aspects of reality, and dismiss it as a misleading form of knowledge. But many times in the past we have realized that it is our immediate intuitions that are imprecise: if we had kept to these we would still believe that the Earth is flat and that it is orbited by the sun. Our intuitions have developed on the basis of